
Rotation-Equivariant Land Cover and Land Use Classification

Milena Perez-Gerus, Sonika Puvvada, Xin Wang

Abstract

Land use and land cover (LULC) classification from satellite imagery often lacks a canonical orientation, yet standard convolutional neural networks (CNNs) are only translation-equivariant and must learn separate filters for rotated versions of the same pattern. Rotation-based data augmentation can help, but it cannot guarantee consistent behavior under rotation. In this work, we examine how architectural rotation equivariance and rotation augmentation each contribute to rotational robustness on the EuroSAT benchmark. We reproduce a shallow CNN baseline and build a C_4 -equivariant counterpart using group convolutions from the E2CNN library, which enforces 90° rotation equivariance and achieves rotation-invariant outputs through group and spatial pooling. We verify the model’s equivariance and invariance numerically and show that implementation details such as padding and pooling choices are critical for preserving symmetry. Under train–test conditions without rotations, the baseline and the equivariant model reach similar accuracy. When both training and testing use rotation augmentation, the baseline nearly matches or slightly exceeds the equivariant model. However, when the models are trained only on upright images and evaluated on rotated ones, the C_4 -equivariant network remains stable, while the baseline drops sharply. An ablation study varying the amount of rotation augmentation during training shows that the baseline becomes rotation-robust only with heavy augmentation, whereas the equivariant model is consistently robust across all conditions. The results demonstrate data augmentation can approximate rotational symmetry, but only equivariant architectures enforce it by design.

1 Introduction

Remote sensing imagery provides critical data for Earth observation tasks such as urban planning [1], agriculture monitoring [2], and environmental assessments [3]. Land use and land cover (LULC) classification is a core task in remote sensing because it assigns semantic labels to regions in satellite imagery to allow for large-scale analysis of landscapes and human activity [4, 5]. Convolutional neural networks (CNNs) achieve strong performance for remote sensing image classification, achieving state-of-the-art performance on tasks such as road extractions, building extractions, and land-cover mapping [6, 7]. EuroSAT [8, 9], derived from Sentinel-2 multispectral imagery [10], serves as a standard benchmark for evaluating CNN land-cover classification models across ten land-cover classes.

However, standard CNNs implicitly assume the features in satellite images occur in a consistent orientation. In practice, overhead imagery does not have a canonical orientation [11]. Patterns such as agricultural fields, road networks, and vegetation frequently appear under arbitrary orientations depending on sensor angle and landscape structure [11]. Since standard CNNs are only equivariant to translations, rotating the input images does not produce a corresponding rotated output image. As a result, CNNs must learn separate filters for different orientations of the same underlying pattern, increasing sample complexity and parameter redundancy [12].

Although data augmentation is commonly used to expose the model to multiple orientations, augmentation alone does not change the underlying architecture and therefore cannot guarantee consistent feature transformations under rotation. In contrast, rotation-equivariant layers encode symmetry directly through weight sharing over the rotation group, providing built-in robustness that augmentation alone cannot achieve [11, 13].

Rotation-equivariant neural networks address this limitation by encoding rotational symmetry directly into the convolutional layers. Although rotation-equivariant networks have been applied to several remote sensing tasks, their effect on the widely used EuroSAT benchmark remains unexplored. In this work, we investigate whether incorporating rotation-equivariance into a CNN architecture improves performance on EuroSAT. We first reproduce the baseline CNN architecture reported in the EuroSAT benchmark and then construct a rotation-equivariant similar architecture using group convolutions [12]. We compare both models under identical training conditions to isolate the contribution of built-in rotation-equivariance to classification performance and robustness.

2 Related Work

Rotation-Equivariant Models in Remote Sensing Lohit and Marks introduce EquiCR, a rotation-equivariant model for satellite cloud removal, where the goal is to reconstruct cloud-free optical images using SAR and partially occluded optical inputs [11]. A central challenge in cloud removal is that satellite content has no canonical orientation, so the reconstructed output should be visually consistent regardless of how the input is rotated. To address this, EquiCR enforces an equivariance constraint: rotating the SAR + cloudy optical input before inference must yield the same output as performing inference first and then rotating the result. This is implemented using group convolutions over the $p4 / p4m$ rotation and reflection groups. The authors show that EquiCR achieves higher reconstruction quality, both qualitatively and quantitatively, than comparable CNNs without built-in equivariance, and the performance gap increases in difficult settings such as small training datasets and high cloud occlusion.

Our work draws on the same architectural mechanism of embedding rotation structure into the model, but applies it to a classification setting rather than reconstruction.

Rotation-Equivariant CNNs for Land Cover Mapping Marcos et al. propose RotEqNet for semantic labeling of high-resolution aerial imagery, where features like buildings, roads, and crop fields appear at arbitrary orientations [14]. The model constructs vector-valued feature responses by applying multiple rotated copies of each filter, which captures both feature magnitude and orientation. A max-pooling step across orientations selects the strongest response and produces compact representations that preserve directional information. This approach encodes fine-grained rotation invariance while sharing filter weights across rotations, reducing parameter count by approximately an order of magnitude relative to standard CNNs. RotEqNet is shown to perform particularly well in limited-label conditions and when models must remain resource-efficient.

Instead of concentrating on vector-field activations and model compression, our emphasis is on adapting group convolutions and evaluating whether rotation-equivariance improves classification generalization under standard benchmark conditions.

Group Equivariant CNN Framework The broader conceptual basis for our approach comes from Group Equivariant Convolutional Networks (G-CNNs), which share filters across transformations belonging to a symmetry group [12]. This produces representations that transform predictably when the input is rotated or reflected, which reduces the need for data augmentation and prevents redundant learning of rotated copies of the same feature. Steerable CNNs extend this approach, providing a systematic way to construct filters that are stable under transformations from planar symmetry groups commonly used in imaging applications [15].

While prior work has demonstrated these frameworks on datasets such as MNIST and CIFAR, their impact on Land Use and Land Cover (LULC) classification has been less systematically evaluated. To fill this gap, we apply group-equivariant architectures to the EuroSAT dataset [8, 9] and perform a controlled comparison against a standard CNN baseline under matched training conditions.

3 Background

CNNs, Translation Equivariance, and Rotation Sensitivity Convolutional Neural Networks (CNNs) are widely used for visual recognition tasks due to their translation equivariance, meaning that shifting the input results in a corresponding shift in feature maps [12]. However, this property does not extend to rotations. A standard CNN fails to connect a rotated pattern with the original pattern it learned. Consequently, the network must dedicate resources to learning every possible rotation of a feature, which increases complexity and can lead to poor performance on rotations not seen during training. This issue is often mitigated through rotation-based data augmentation, which does not enforce exact rotation consistency and still requires redundant learning of rotated variants. Thus, augmentation approximates robustness rather than guaranteeing it.

Equivariance and Invariance A more structured approach is to design models that reflect the inherent geometric structure of the data. A model Φ is said to be equivariant to a transformation g if $\Phi(gx) = g\Phi(x)$, meaning that applying a transformation to the input results in the same transformation being applied to the internal representation [12]. For classification tasks, we typically want equivariant intermediate representations and a final prediction that is invariant, meaning it does not change under transformation. In the context of LULC classification, this means that internal features should rotate when the input rotates, but the final class label should remain the same.

Discrete Rotation Symmetry Groups The possible rotations of square-grid images can be described using symmetry groups. The finite cyclic group C_4 consists of the four discrete in-plane rotations by 0° , 90° , 180° , and 270° , capturing rotation symmetry at the feature level. The wallpaper group $p4$ contains the same four right-angle rotations but additionally includes planar translations, making it a symmetry group of the entire image domain. The $p4m$ group extends $p4$ by further including mirror reflections. These groups are appropriate for overhead satellite images, where rotational symmetry is present but continuous rotation invariance is not necessary for effective modeling [12].

Group Convolutions Group Convolutional Neural Networks (Group CNNs) extend standard CNNs by sharing filters across a wider class of transformations represented by a symmetry group H , rather than only translations. This means that one learned filter can detect the same pattern across multiple orientations. For an input feature field $f : H \rightarrow \mathbb{R}^{c_{in}}$ and a group filter $\Psi : H \rightarrow \mathbb{R}^{c_{in} \times c_{out}}$, the group convolution is defined as

$$[f * \Psi](g) = \sum_{h \in H} f(h) \Psi(g^{-1}h),$$

where g and h index orientation elements of H (such as the discrete rotation group C_4). This construction ensures that the resulting feature maps transform coherently with the input, yielding built-in rotation equivariance. The output therefore contains an ‘‘orientation channel,’’ which encodes both the presence of a feature and the direction in which it appears.

Group Pooling and Rotation-Invariant Outputs To obtain rotation-invariant predictions from equivariant features, the orientation dimension indexed by H is removed by applying a group pooling operator, such as summation or averaging over all $g \in H$. This produces a representation invariant under the action of H , allowing the model’s final prediction to remain identical for any rotated version of the input.

Steerable CNN Framework and Problem Setting Steerable CNNs [15] provide a general framework for constructing group-equivariant architectures. The symmetry acting on images is the roto-translation group $G = \langle H, (\mathbb{R}^2, +) \rangle$, where $H \leq O(2)$ encodes discrete rotations and reflections. Feature fields in each layer transform according to group representations $\rho_{in} : H \rightarrow \text{GL}(\mathbb{R}^{c_{in}})$ and $\rho_{out} : H \rightarrow \text{GL}(\mathbb{R}^{c_{out}})$, which specify how feature channels rotate or reflect. Group convolutions enforce the equivariance condition

$$\Phi_\theta(\rho_{in}(g)f) = \rho_{out}(g)\Phi_\theta(f), \quad \forall g \in H,$$

and a final pooling operation over H produces rotation-invariant descriptors.

In our application to land cover and land use classification on the EuroSAT dataset, an image is represented as $f : \mathbb{Z}^2 \rightarrow \mathbb{R}^3$, and the task is to learn a mapping $\Phi_\theta : f \mapsto y$ with $y \in \{1, \dots, 10\}$. We adopt the discrete rotation subgroup $H = C_4 \leq SO(2)$, representing 90° rotations, and the

corresponding roto-translation group $G = \langle C_4, (\mathbb{R}^2, +) \rangle$, often denoted $p4$. Our objective is to learn a model Φ_θ that is equivariant with respect to the action of C_4 , ensuring that internal feature fields rotate consistently while final predictions remain invariant.

4 Methods

4.1 Baseline Architecture

We modify the shallow CNN architecture from the EuroSAT [8, 9] paper for the native image resolution and without ImageNet based preprocessing. The baseline model consists of three convolutional blocks, each with a 3×3 convolution (stride 1, padding 1), ReLU activation, and 4×4 max pooling with stride 2. The resulting features are flattened and passed to a fully connected classifier.

4.2 C_4 -Equivariant Architecture

To assess whether explicit architectural rotation equivariance can improve robustness to unseen orientations, we plan to replace conventional convolutions with group convolutions over the C_4 group, corresponding to discrete rotations of 0° , 90° , 180° , and 270° .

Our model is a rotation-equivariant shallow CNN built using E2CNN library, trained under the same hyperparameter settings as the baseline models, and designed to enforce symmetry under cyclic rotations. This will allow direct comparison between data-level augmentation and architecture-level symmetry handling. It begins by constructing the appropriate gspace and FieldTypes, encoding inputs with trivial representations and hidden layers with regular representations so that each convolution layer is fully equivariant to the chosen group C_4 . The network contains three equivariant convolution blocks—each consisting of an R2Conv, a nonlinearity ReLU, and spatial max-pooling—followed by group pooling, which converts the final C_4 -equivariant feature maps shown in Figure 1 into C_4 -invariant representations. A final linear classifier maps these invariant features to output class scores. Overall, the architecture preserves symmetry throughout the feature extractor and guarantees rotation invariance at the classifier.

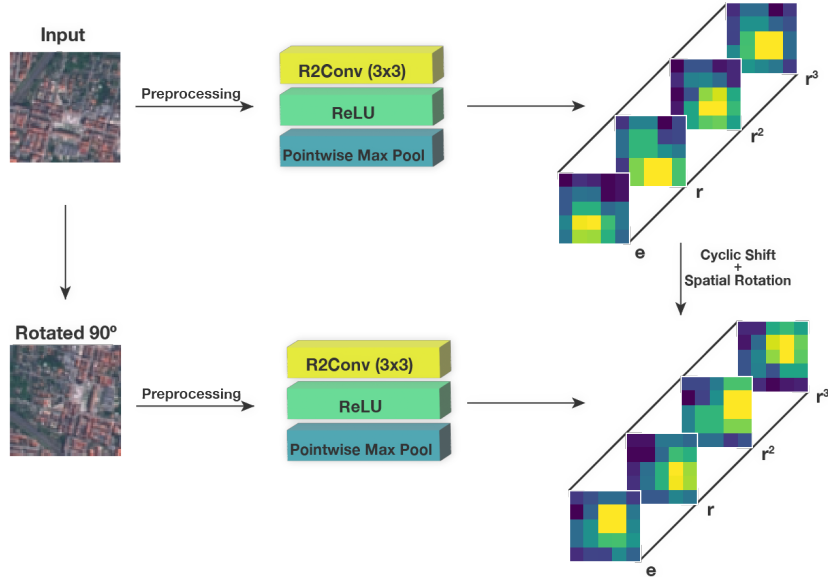


Figure 1: When the image is rotated by 90° , the R2Conv block produces feature maps that rotate accordingly and shift cyclically across orientation channels, visualizing the built-in C_4 -equivariance of the network.

As part of model validation, we conducted both equivariance and invariance tests to verify correct implementation of the C_4 symmetry constraints. Our corrected model passed all tests with numerical

errors below 10^{-5} . The testing protocol is described in Appendix B, and quantitative/qualitative results are provided in Sec. 6.1.

5 Experimental Setup

5.1 Data Preprocessing

Following the EuroSAT [8, 9] protocol, the dataset was randomly split into 80% for training and 20% for testing. Unlike the original EuroSAT paper, which resized all images to 224×224 for compatibility with large models such as GoogLeNet [16] and ResNet-50 [17, 18], we keep all RGB patches at their native resolution of 64×64 . This reduces training time and enables controlled comparisons between the baseline and C_4 -equivariant architectures. All images are normalized using EuroSAT RGB channel statistics rather than ImageNet statistics.

To evaluate architectural rotation equivariance, we restrict data augmentation to only discrete C_4 rotations. Rotation augmentation is performed using `torch.rot90`, which applies exact tensor permutations. No additional geometric transformations are used to ensure that the symmetry of the training distribution matches the C_4 group implemented in the equivariant network. Rotation augmentation is applied only within specific experimental settings described in Section 5.3.

5.2 Training Details

The network is trained from scratch using stochastic gradient descent (learning rate 10^{-3} , momentum 0.9, batch size 16) and cross-entropy loss. To allow for direct comparison with the equivariant architecture, all experiments share identical optimization settings and differ only in the rotation augmentation applied during training.

Although the original EuroSAT shallow CNN was trained for 120 epochs, loss curves in our setting showed that both the baseline and the C_4 -equivariant model converged by approximately 65 epochs under all augmentation conditions. To reduce computational time and enable consistent comparisons across multiple experiments, we report all results at 65 training epochs. Training models for 120 epochs yielded no substantial improvements in test accuracy, which confirmed that extending training does not change the conclusions of our study and 65 epochs were sufficient for convergence.

5.3 Datasets and Tasks

All experiments are conducted on the EuroSAT RGB land-cover dataset, consisting of 27,000 Sentinel-2 image patches across 10 classes (e.g. Residential, Annual Crop, Forest, Highway). Each image is 64×64 pixels. Consistent with our preprocessing described in Section 5.1, images are normalized and split into an 80/20 train-test partition. Full class distributions and example images are provided in Appendix A.

The task is 10-way land-cover classification. To evaluate how rotation affects model generalization, we structure three train-test conditions that differ in the presence or absence of rotation-based augmentation:

Experiment 1: Train (No Augmentation) $\rightarrow$ Test (No Augmentation) This experiment establishes a baseline for the shallow CNN and the C_4 -equivariant CNN when the dataset contains no rotational variation at either stage. We do not expect meaningful differences between the models in this setting since the distribution is aligned across train and test.

Experiment 2: Train (Rotation Augmentation) $\rightarrow$ Test (Rotation Augmentation) To study how augmentation interacts with the architectural invariance, both models are trained with rotation-only augmentation and evaluated on similarly augmented data. We focus on discrete rotations (0° , 90° , 180° , 270°). For this milestone, we fix a single augmentation rate; tuning will be explored in the final report.

Experiment 3: Train (No Augmentation) $\rightarrow$ Test (Rotation Augmentation) To measure rotational generalization, models are trained only on upright images but evaluated on rotated inputs.

Standard CNNs typically degrade sharply here, while the C_4 model should maintain stable performance. This experiment is therefore designed to reveal the largest expected gap between the two architectures.

Ablation Study: Probabilistic Rotation Sampling For the ablation study in Section 5, we vary the proportion of rotated samples seen during training. Let $p_{\text{train}} \in \{0.1, 0.50, 1.00\}$ denote the probability that a training sample is rotated. When a rotation is applied, the angle is sampled uniformly from the C_4 group, i.e., $\{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$. The corresponding test set is fixed at $p_{\text{test}} = 0.5$. This procedure allows us to evaluate whether increased rotation exposure during training can approximate the built-in equivariance of the C_4 network.

5.4 Metrics

We train all models using the cross-entropy loss and monitor training loss per epoch to ensure convergence. Overall classification accuracy serves as the primary evaluation metric across augmentation settings. To assess robustness to C_4 rotations, we measure the discrepancy between the network responses to x and to transformed inputs gx using the ℓ_∞ norm: $\varepsilon_\infty(h; x, g) = \|h(x) - h(gx)\|_\infty$. Small values indicate stable behavior under group actions: when $h(\cdot)$ is taken from intermediate layers, ε_∞ reflects equivariance consistency of feature maps, and when taken from classifier outputs, it measures rotation invariance of the predictions.

6 Results

6.1 Verifying Equivariance and Invariance, and Architectural Corrections

A C_4 -equivariant network must satisfy the property $\Phi(gx) = g\Phi(x)$, $g \in C_4$, meaning that rotating the input by any element of the cyclic group C_4 should produce a correspondingly rotated feature representation. To ensure that our implemented C_4 -equivariant CNN satisfies this property, we performed controlled equivariance and invariance tests on untrained and trained versions of the model. The tests follow the procedure illustrated in Appendix B.

Initial Equivariance Failures. Our first model version violated equivariance substantially, with mean errors exceeding 0.07 between $\Phi(gx)$ and $g\Phi(x)$. Visualizations showed clear mismatches across the four C_4 orientations. This indicated that the architecture contained operations that did *not* respect the rotational symmetry of the group.

Identifying Sources of Equivariance Breaking. Through systematic ablations, we found two main causes: **(1) Non-equivariant padding.** Standard zero-padding introduces boundary patterns that do not transform equivariantly under 90° rotations. Removing all padding from the equivariant convolution blocks reduced equivariance errors by several orders of magnitude. **(2) Pooling with incompatible kernel sizes.** In the original model architecture design, a 4×4 max-pooling kernel breaks equivariance because its sampling grid is not preserved under C_4 rotations. Replacing it with a 3×3 kernel restored equivariant behavior. This aligns with the steerable CNN requirement that all spatial operations must be symmetric under the same group action.

After applying these corrections, the feature extractor achieved near-perfect equivariance both before and after training, with maximum errors on the order of 10^{-6} .

As shown in Figure 3, the transformed feature maps $g\Phi(x)$ align almost perfectly with $\Phi(gx)$ across all four rotation elements, confirming that the corrected architecture enforces the desired C_4 -equivariance.

Invariance Testing and Output-Level Correction. Although the corrected model produced equivariant feature maps, the *classifier outputs* were not rotation-invariant; rotating the input still changed the predicted logits. This is expected because group pooling alone removes only the *orientation* dimension but preserves spatial variability. Thus, even after C_4 group pooling, the features retain a spatial grid of shape (B, C, H, W) .

For classification, we require

equivariant features $\xrightarrow{\text{group pooling}}$ orientation-invariant features $\xrightarrow{\text{spatial pooling}}$ rotation-invariant output.

Following the steerable CNN formulation, we added a final **global spatial average pooling** layer after group pooling which produces invariant descriptors of shape $(B, C, 1, 1)$.

After introducing this final step, invariance errors decreased to numerical tolerance ($< 10^{-5}$), and the model achieved 93.91% accuracy on EuroSAT (64×64 resolution, 65 epochs). Figure 4 visualizes this result: although the orientation-indexed feature fields change with input rotation, the group-pooled and spatially-averaged representations remain identical, yielding a strictly rotation-invariant descriptor for classification.

6.2 Experiment Results

Experiment 1: No Augmentation $\rightarrow$ No Augmentation This setting evaluates both models when the train and test distributions contain upright, unrotated imagery. Because no rotational variability is introduced, we expect similar performance from both architectures. As shown in Table 1, the baseline CNN and the C_4 -equivariant model achieve nearly identical accuracy (93.04% vs. 93.74%), confirming that explicit symmetry does not provide a benefit when the data distribution is already aligned.

Experiment 2: Rotation Augmentation $\rightarrow$ Rotation Augmentation Here, both training and testing apply discrete C_4 rotations ($0^\circ, 90^\circ, 180^\circ, 270^\circ$). If rotation augmentation alone can approximate invariance, the baseline CNN may be able to close the gap typically associated with equivariant networks. Table 1 shows that both models improve, but the baseline CNN benefits more (95.69% vs. 94.20%). This suggests that exposure to rotated samples allows a standard CNN to learn orientation-agnostic features, reducing the advantage of built-in symmetry.

Experiment 3: No Augmentation $\rightarrow$ Rotation Augmentation This experiment isolates rotational generalization by training only on upright images and testing on rotated ones. Unlike the baseline, the equivariant model does not rely on augmented samples to handle unseen orientations. Consistent with this expectation, the baseline CNN suffers a large drop in accuracy (from 93.04% to 90.09%), while the C_4 -equivariant model remains robust and even improves relative to Experiment 1 (92.98% to 94.00%). These results show that built-in equivariance provides genuine rotation robustness without requiring additional training data.

Exp.	Train Aug.	Test Aug.	Baseline Acc.	C_4 -Equiv. Acc.
1	None	None	93.04%	93.74%
2	C_4 Rot.	C_4 Rot.	95.69%	94.20%
3	None	C_4 Rot.	90.09%	94.00%

Table 1: Test accuracy under three train–test rotation conditions. Only the C_4 -equivariant model maintains high accuracy without rotation augmentation, indicating robustness to unseen orientations.

Ablation Study: Varying Train-Time Rotation Exposure To further isolate the interaction between augmentation and architectural symmetry, we vary the probability of applying rotations during training (p_{train}) while keeping the test augmentation fixed. This comparison evaluates whether increasing rotational diversity during training can approximate the invariance induced by the equivariant architecture.

We study three conditions:

- Train < Test: $p_{\text{train}} = 0.10$
- Train = Test: $p_{\text{train}} = 0.50$
- Train > Test: $p_{\text{train}} = 1.00$

Table 2 shows that the baseline CNN only becomes rotation-robust when heavily exposed to rotated samples. Meanwhile, the C_4 -equivariant model maintains stable accuracy regardless of training

exposure, indicating that architectural symmetry inherently provides robustness without tuning augmentation.

Condition	p_{train}	Baseline Acc	C_4 -Equiv. Acc
Train < Test	0.10	94.09%	94.37%
Train = Test	0.50	95.33%	94.24%
Train > Test	1.00	95.50%	94.39%

Table 2: Accuracy scores on ablation results varying the proportion of rotated samples during training. The equivariant model remains robust without extensive augmentation, while the baseline requires heavy augmentation to approximate rotational invariance.

7 Discussion

Our results highlight when rotation augmentation can substitute for equivariance and when it cannot. When train and test images share a canonical orientation (Exp. 1), both models perform comparably, with only a slight improvement of 0.7% in the C_4 -equivariant model, indicating that rotational symmetry provides no benefit when no variability is present. When both training and testing include discrete C_4 rotations (Exp. 2), the baseline CNN nearly matches the equivariant model, suggesting that a shallow network can approximate rotation robustness once it observes enough rotated examples. However, when evaluated on unseen orientations without rotation exposure (Exp. 3), the baseline accuracy drops while the C_4 model remains stable, demonstrating that only architectural symmetry guarantees robustness without augmented data.

The ablation study further shows diminishing returns from augmentation. Once the baseline has seen a minimal set of rotated examples, additional exposure yields little improvement, while the equivariant network remains unaffected across all augmentation rates. This indicates that augmentation can encourage approximate rotation invariance in the final predictions, but it does not enforce symmetry in the feature representations themselves. By contrast, the C_4 model achieves invariance through equivariance by guaranteeing rotation-consistent features that become invariant after group pooling. Our equivariance and invariance tests confirm this architectural guarantee and also empirically show that implementation details (e.g., padding and kernel choices) must respect the group action to preserve symmetry. However, the precise mechanism by which zero-padding disrupts equivariance remains unclear and warrants further investigation in future work.

The small performance gaps on EuroSAT are possibly due to the fact that many land-cover classes are distinguishable by spectral or textural cues rather than orientation-specific geometry, reducing the utility of explicit rotation symmetry. Future work should evaluate equivariant models on tasks with greater orientation dependence within remote sensing and explore extensions beyond discrete C_4 rotations such as continuous $SO(2)$ symmetry and settings with limited labeled data, where inductive biases are expected to provide greater gains.

8 Conclusion

We investigated how rotation augmentation and architectural equivariance each contribute to rotational robustness in land-cover classification. Our experiments show that a C_4 -equivariant CNN consistently generalizes to unseen orientations, while a standard CNN can only approximate this robustness when exposed to rotated examples during training. When rotational variability is matched across train and test, augmentation allows the baseline to nearly match equivariant performance, but this approximation offers diminishing gains and does not guarantee invariance. These findings suggest that data augmentation can imitate symmetry yet only equivariant design enforces it. Future work should examine remote-sensing tasks where object identity depends more strongly on geometric orientation, and extend beyond discrete C_4 rotations toward continuous $SO(2)$ symmetry and limited-label settings.

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A EuroSAT Dataset Details

The EuroSAT dataset [9] is a land-use and land-cover classification benchmark derived from Sentinel-2 satellite imagery. Each image is RGB with spatial resolution 64×64 pixels and belongs to one of ten classes: *AnnualCrop*, *Forest*, *HerbaceousVegetation*, *Highway*, *Industrial*, *Pasture*, *PermanentCrop*, *Residential*, *River*, and *SeaLake*. Each class contains 2000–3000 samples, totaling 27,000 images.

Train–Test Split. We use an 80%–20% stratified split across all classes. Table 3 reports the number of images per class in the original dataset and in each split.

Table 3: Number of EuroSAT images per class before and after splitting.

	Annual	Forest	HerbVeg	Highway	Indust.	Pasture
Original	3000	3000	3000	2500	2500	2000
Train (80%)	2400	2400	2400	2000	2000	1600
Test (20%)	600	600	600	500	500	400

	PermCrop	Res.	River	SeaLake	Total
Original	2500	3000	2500	3000	27000
Train (80%)	2000	2400	2000	2400	21600
Test (20%)	500	600	500	600	5400

Example Images. Figure 2 shows one representative image per class. These examples illustrate the visual diversity of the dataset, including vegetation types, agricultural patterns, urban structures, and natural water bodies.



Figure 2: Example images from the ten EuroSAT classes.

B Equivariance and Invariance Testing Procedures

In this section, we describe the procedures used to verify both (1) rotation equivariance of the intermediate feature maps and (2) rotation invariance of the final classifier output in our C_4 -equivariant CNN.

B.1 Equivariance Test

To verify equivariance, we evaluate whether the feature extractor Φ satisfies

$$g \cdot \Phi(x) = \Phi(g \cdot x),$$

for every rotation g in the cyclic group C_4 .

Numeric Equivariance Test. Given an input image x , we compute:

$$\begin{aligned} y_1 &= \Phi(x), \\ y_1^g &= g \cdot y_1, \\ y_2 &= \Phi(g \cdot x). \end{aligned}$$

The maximal absolute difference

$$\Delta_{\text{equiv}} = \|y_1^g - y_2\|_{\infty}$$

is used as the equivariance error. In our experiments, this error remains below 10^{-5} across all group elements, indicating excellent numerical equivariance.

Feature Map Visualization. We further reshape the output into feature fields (each consisting of four channels corresponding to the C_4 elements) and visualize:

$$\Phi(x), \quad g \cdot \Phi(x), \quad \Phi(g \cdot x).$$

The resulting fields exhibit cyclic permutation under rotation as shown in Figure 3, demonstrating that the learned representations transform according to the group action.

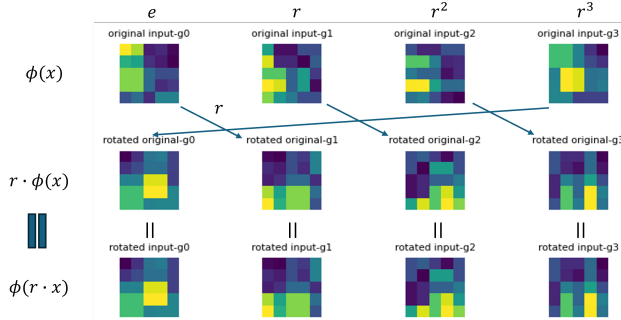


Figure 3: Equivariance test for the corrected model. The first row shows the feature maps $\Phi(x)$ for the four group elements g_0, g_1, g_2, g_3 . The second row shows $g \Phi(x)$ obtained by rotating the feature maps themselves, and the third row shows $\Phi(gx)$ obtained by feeding the rotated inputs into the network. For a perfectly equivariant model, $g \Phi(x)$ and $\Phi(gx)$ coincide for every $g \in C_4$, which is visually confirmed by the alignment between the second and third rows.

B.2 Invariance Test

After applying group pooling, the network implements an invariant mapping

$$\Psi(x) = \Psi(g \cdot x), \quad \forall g \in C_4.$$

Numeric Invariance Test. For each rotated input $x_g = g \cdot x$, we compute the classifier logits:

$$\ell_g = \Psi(x_g).$$

We measure the maximal deviation relative to the reference logit ℓ_e :

$$\Delta_{\text{inv}} = \max_{g \in C_4} \|\ell_g - \ell_e\|_{\infty}.$$

Across all experiments, $\Delta_{\text{inv}} < 10^{-5}$, confirming stable invariance.

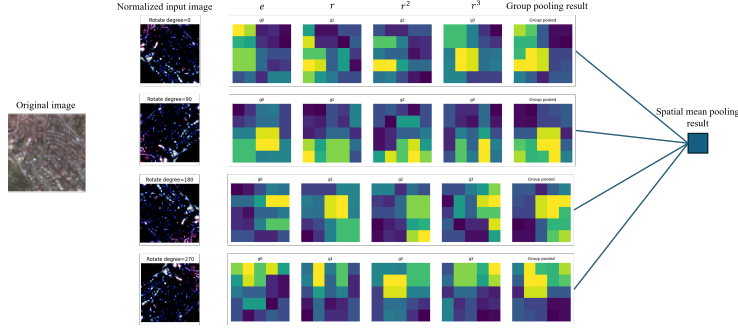


Figure 4: Invariance test for the final model. Each row corresponds to an input image rotated by 0° , 90° , 180° , or 270° . For each rotated input, we visualize the four orientation-indexed feature fields (g_0 – g_3) produced by the equivariant feature extractor, followed by the group-pooled representation that removes the orientation dimension. All group-pooled maps remain identical across rotations. After applying spatial global pooling (right), the final representation used by the classifier is exactly rotation-invariant, matching the behavior measured in our quantitative tests.

Visualization of Pooled Features. We additionally visualize the pooled representations for all rotated versions of the image in Figure 4. As expected, all rotations produce nearly identical results, reflecting the invariance induced by spatial and group pooling.

B.3 Summary

Both the numeric evaluations and the visualization results confirm that:

- before pooling, the network is **rotation-equivariant**, and
- after pooling, the network becomes **rotation-invariant**.

These tests validate the correctness of our C_4 -invariant CNN implementation.

B.4 Implementation Notes

- All group actions g_k use rotation operators from `e2cnn`.
- Feature comparisons are performed on tensors *before* the classifier for equivariance, and *after* spatial global pooling for invariance.
- Tests are first run on untrained models to verify architectural correctness, then repeated after training.